

Nate Cirino

832-833-5202 | cirino.na@northeastern.edu | [portfolio](#) | [linkedin.com/in/nate-cirino](https://www.linkedin.com/in/nate-cirino) | github.com/nateci

Graduating May 2027 | Available to start 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Algorithms & Data Structures, Computer Systems, Computer Networks, Digital Design & Computer Architecture, Foundations of Data Science, Object-Oriented Design, Foundations of Software Engineering, Systems Security, Cybersecurity

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches
- Own front-of-house device stack across all Wayfair retail brands (AllModern, Birch Lane, Joss & Main, Perigold)
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Optimized Expert AI query understanding, search, and filtering across 20+ microservices, improving retrieval relevance for clinical queries at scale
- Integrated Expert AI into CME accrual/redemption microservices, scaling GenAI decision support to 250K+ clinicians
- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts under load and improving end-to-end throughput
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Flask backend with an end-to-end RAG indexing and retrieval pipeline serving personalized recommendations
- Evolved the retrieval stack from LangChain to a Vespa search engine to zero-ops Bedrock KB, improving result quality and cutting serving overhead
- Sustained 99.99% uptime across 1–2M monthly LLM inference calls for 2K paying users via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup into DevOps, dev, and agentic-workflow teams; continue as technical advisor

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and offline data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI SRE agent that autonomously triages incidents through multi-step tool-calling over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG index over runbooks and postmortems with prompt caching on frozen schemas, cutting inference cost; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and an LSM-tree storage engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTable indexes
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference acceleration under hard real-time latency limits

SKILLS & HONORS

Languages/Tools: Python, C++, C, Java, Rust, TypeScript, JavaScript, Bash, Linux, Docker, Kubernetes, AWS, Azure, PostgreSQL, GraphQL, Terraform, Jenkins, Buildkite, Groovy

Frameworks & Libraries: PyTorch, TensorFlow, Pandas, FastAPI, Flask, LangChain, Vespa, pgvector, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor